

LMS Platform Expansion Blueprint

Architecture, Build vs. Buy Strategy, Infrastructure & Third-Party Pricing Breakdown for Batch-wise Live Classes, Video Lectures & E-Library

CURRENT STACK	TARGET CAPACITY	EST. DEV TIMELINE	EST. MONTHLY INFRA
Node.js • Express • MySQL • React	500–5,000 Concurrent Students	8 – 11 Weeks (Phased)	\$90 – \$130 / mo (Starter Scale)

1 Scope & Feature Expansion Overview

🖥️ Live Classes (Batch-wise)

Interactive real-time audio/video streaming with built-in engagement tools:

- **WebRTC SFU Grid:** Ultra-low latency video/audio (<200ms)
- **Interactive Tools:** In-class Chat, Whiteboard, Live Polls & Raise Hand
- **Auto Cloud Recording:** Composite MP4 output direct to batch archives

🎥 Video Lectures (Batch-wise)

Pre-recorded lessons with seamless playback and piracy prevention:

- **Adaptive Bitrate (HLS):** Auto multi-quality (1080p, 720p, 480p, 360p)
- **Dynamic Watermark:** Moving student name/phone overlay to stop leaks
- **Direct-to-Cloud Upload:** Fast S3/R2 direct uploads with presigned URLs

📖 Study Materials (E-Library)

Structured digital repository organized by Batch, Subject & Chapter:

- **Protected Reader:** In-browser PDF renderer (React-PDF/PDF.js)
- **Security Controls:** Block right-click, print, and direct download links
- **Zero Egress CDN:** Fast global document distribution via Cloudflare R2

2 Build from Scratch vs. 3rd-Party Decision Matrix

MODULE / COMPONENT	FEASIBILITY OF SCRATCH BUILD	RECOMMENDED TOOL / PROVIDER	STRATEGIC RATIONALE & TRADE-OFFS
Live Video SFU Audio/Video Multiplexing	❌ Never Build Scratch	LiveKit / 100ms / Agora	Writing C++/Rust SFU, global STUN/TURN, and jitter buffers takes 12+ months of R&D. Use ready SFU SDKs.
Cloud Live Recording Composite Video + Board	❌ High DevOps Burden	LiveKit Egress / Agora Cloud Rec	Requires automated headless browser cluster to record layout and output MP4. Cloud egress does this automatically.
Interactive Whiteboard Drawing & Equation Canvas	⚠️ Hybrid (Lib + Custom Sync)	Excalidraw / Fabric.js + Socket.io	Embed open-source canvas UI; sync vector delta JSON over existing Node.js WebSocket. Zero license fees.
Live In-Class Polls & Chat MCQs, Timers & Messaging	✅ 100% In-House Build	Node.js + Socket.io + Redis	Easy to build in-house on current Express stack; zero per-message SaaS fees.
Video Transcoding & HLS VOD Multi-bitrate Streaming	⚠️ Use Managed Video CDN	Bunny.net Stream / Cloudflare Stream	Avoid configuring manual FFmpeg workers. Managed video CDN encodes, hosts, and streams for pennies.

MODULE / COMPONENT	FEASIBILITY OF SCRATCH BUILD	RECOMMENDED TOOL / PROVIDER	STRATEGIC RATIONALE & TRADE-OFFS
File Storage & CDN PDFs, Notes, Uploads	 Cloud Infra	Cloudflare R2 + CDN	\$0 bandwidth egress fees; eliminates surprise bills when thousands download 50MB PDFs.
PDF Viewer & Watermark E-Library Reader	 100% In-House Build	React-PDF / PDF.js + Canvas	Custom UI with user watermarking & download prevention. Completely free & proprietary.

 **Core Architecture Philosophy**
Offload **heavy media networking** (SFU & Video Transcoding) to specialized low-cost cloud providers, while building all **business logic, access controls, interactive polls, chat, and document readers** 100% in-house in your Node.js + React codebase.

Technical Architecture & Database Schema

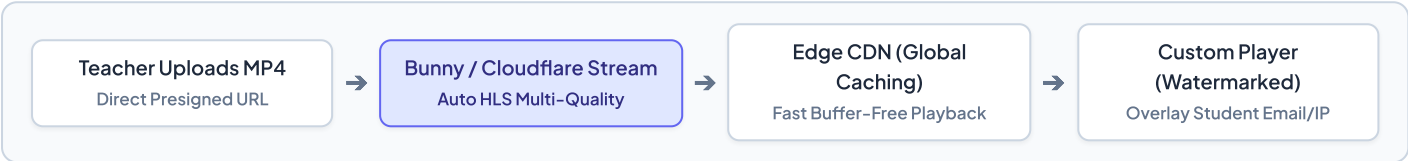
Relational Data Model & Video/Live Pipeline Workflows

3 End-to-End System Pipelines

A. Live Class Architecture (WebRTC + Realtime Layer)



B. Video Lecture Upload & Transcoding Pipeline (VOD)



4 MySQL Schema Extensions (Batch-Integrated)

```
-- 1. Live Classes Table (Integrated with Existing Batches) CREATE TABLE live_classes ( id VARCHAR(36) PRIMARY KEY, batch_id VARCHAR(36) NOT NULL, title VARCHAR(255) NOT NULL, instructor_id VARCHAR(36) NOT NULL, scheduled_start_time DATETIME NOT NULL, status ENUM('SCHEDULED', 'LIVE', 'COMPLETED', 'CANCELLED') DEFAULT 'SCHEDULED', room_sid VARCHAR(255) NULL, -- LiveKit / Agora Room Identifier recording_url VARCHAR(512) NULL, created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP, FOREIGN KEY (batch_id) REFERENCES batches(id) ON DELETE CASCADE ); -- 2. Pre-recorded Video Lectures CREATE TABLE video_lectures ( id VARCHAR(36) PRIMARY KEY, batch_id VARCHAR(36) NOT NULL, subject VARCHAR(100) NOT NULL, title VARCHAR(255) NOT NULL, hls_playlist_url VARCHAR(512), -- CDN .m3u8 Playlist duration_seconds INT DEFAULT 0, transcoding_status ENUM('PENDING', 'PROCESSING', 'READY', 'FAILED') DEFAULT 'PENDING', FOREIGN KEY (batch_id) REFERENCES batches(id) ON DELETE CASCADE ); -- 3. Study Materials & E-Library Documents CREATE TABLE study_materials ( id VARCHAR(36) PRIMARY KEY, batch_id VARCHAR(36) NOT NULL, subject VARCHAR(100) NOT NULL, category ENUM('NOTES', 'ASSIGNMENT', 'PYQ', 'BOOK') NOT NULL, title VARCHAR(255) NOT NULL, file_url VARCHAR(512) NOT NULL, is_protected BOOLEAN DEFAULT TRUE, -- Dynamic In-App Watermarking FOREIGN KEY (batch_id) REFERENCES batches(id) ON DELETE CASCADE );
```

5 Implementation Effort & Phased Roadmap

Phase & Module	Key Development Tasks	Complexity	Effort (1 Fullstack Dev)
Phase 1: E-Library	Cloudflare R2 Direct Uploads + Batch Catalog + React-PDF with Watermark	Low	1.5 – 2.0 Weeks
Phase 2: Video Lectures	Bunny/Cloudflare Stream Hook + Video.js Player + Watch Analytics & Security	Medium	2.5 – 3.0 Weeks
Phase 3: Live Classes (Core)	LiveKit Video Room Grid + Host Controls (Mute All) + Socket.io In-Class Chat	High	3.0 – 4.0 Weeks
Phase 4: Live Interactivity	Collaborative Whiteboard delta sync + In-Class Polls + Auto Cloud Recording	Medium	2.0 – 2.5 Weeks
Total Project Scope	Complete LMS Expansion (Live + VOD + Materials + Analytics)	—	~8 – 11 Weeks

Comprehensive Third-Party Pricing Comparison

Transparent Cost Analysis & Real-World Monthly Budget Simulation

6 Vendor Pricing Comparisons

A. Live Video Streaming & SFU (Live Classes)

Provider	Free Tier	Participant Video Rate	Cloud Recording	Self-Hosting?	Evaluation Verdict
LiveKit Cloud	10,000 min/mo	\$0.0035 / min (720p)	\$0.015 / min	Yes (Open Source)	🏆 Best Pick: Modern SDK, zero lock-in
100ms.live	10,000 min/mo	\$0.0040 / min	\$0.015 / min	No	🕒 Fast UI, but proprietary SaaS
Agora.io	10,000 min/mo	\$0.00399 / min	\$0.0135 / min	No	🏢 Established enterprise; complex APIs
Self-Hosted LiveKit (VPS)	Unlimited	\$0 / min (Server cost only)	Free on VM worker	100% Dedicated	🏆 Cheapest at 2k+ scale: ~\$40/mo VM

B. Video on Demand (VOD Lectures) Hosting & Transcoding

Provider	Transcoding / Encoding	Storage Cost	CDN Bandwidth Cost	Built-in Player & Security
Bunny.net Stream	FREE / \$0	\$0.01 / GB / mo	\$0.005 / GB	✅ HLS Player, DRM Token, Dynamic Watermark
Cloudflare Stream	FREE	\$5.00 / 1,000 min (\$0.005/min)	\$1.00 / 1,000 min viewed	✅ Global Anycast Network, Secure Embed
AWS S3 + MediaConvert	\$0.015 / min encoded	\$0.023 / GB	\$0.085 / GB (CloudFront)	❌ High setup complexity, expensive bandwidth

C. Document & PDF Storage (E-Library)

Storage Provider	Free Tier	Storage Price	Bandwidth Egress (Downloads)	Recommendation
Cloudflare R2	10 GB / month	\$0.015 / GB	\$0.00 (Zero Egress Fees)	🏆 Must-Have: 100% immune to download spikes
AWS S3	5 GB (1st yr)	\$0.023 / GB	\$0.085 / GB	⚠️ Risky for high student PDF downloads

7 Monthly Cost Simulation (500 Active Students Benchmark)

Workload Assumptions: 500 Students, 40 Hours Live Classes/mo (avg 50/class), 100 Hrs VOD Catalog (5,000 hrs watched), 100 GB PDF E-Library.

Category	Chosen Infrastructure / Provider	Calculated Monthly Bill
Live Video SFU	LiveKit Cloud (Using Free Tier + Overage) or Hetzner VM	\$40.00 – \$70.00 / mo
Live Class Cloud Recording	LiveKit Egress → Directly written to Cloudflare R2	\$35.00 / mo (40 hrs @ \$0.015/min)
Video Lectures (Transcoding + CDN)	Bunny.net Stream (100GB Storage + 3TB Bandwidth)	\$16.00 / mo
Study Material Notes (E-Library)	Cloudflare R2 (100GB Storage + \$0 Unlimited Egress)	\$1.50 / mo

CATEGORY	CHOSEN INFRASTRUCTURE / PROVIDER	CALCULATED MONTHLY BILL
Chat, Live Polls & Whiteboard Sync	Self-Hosted Redis on Existing Node.js VM	\$0.00 / mo

Total Estimated Third-Party Operating Cost
For complete Live + VOD + E-Library scale (500 Students)

~\$92 – \$122 / Month

 **Key Takeaway for Team:** You can launch enterprise-grade Live Classes, Video Streaming, and an E-Library with **under \$120/month** in third-party services and **zero upfront server licensing fees**.